

200G Optical Compute Interconnect Line Interface Specification

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Revision History

Revision	Date	Editors	Changes
v1.0	03/11/2026	AMD, Broadcom & Meta	1. Release version

Introduction

This document defines the technical specifications for the line side optical interface of optical compute interconnect (OCI) physical layer. OCI aims to use a dense wavelength-division multiplexing (DWDM) wavelength grid with cascaded micro-ring resonators (MRR) to enable a low-power high-density solution for artificial intelligence (AI) backend network optical interconnects.

An illustration of a cross-section view of different OCI implementations is shown in Figure 1.

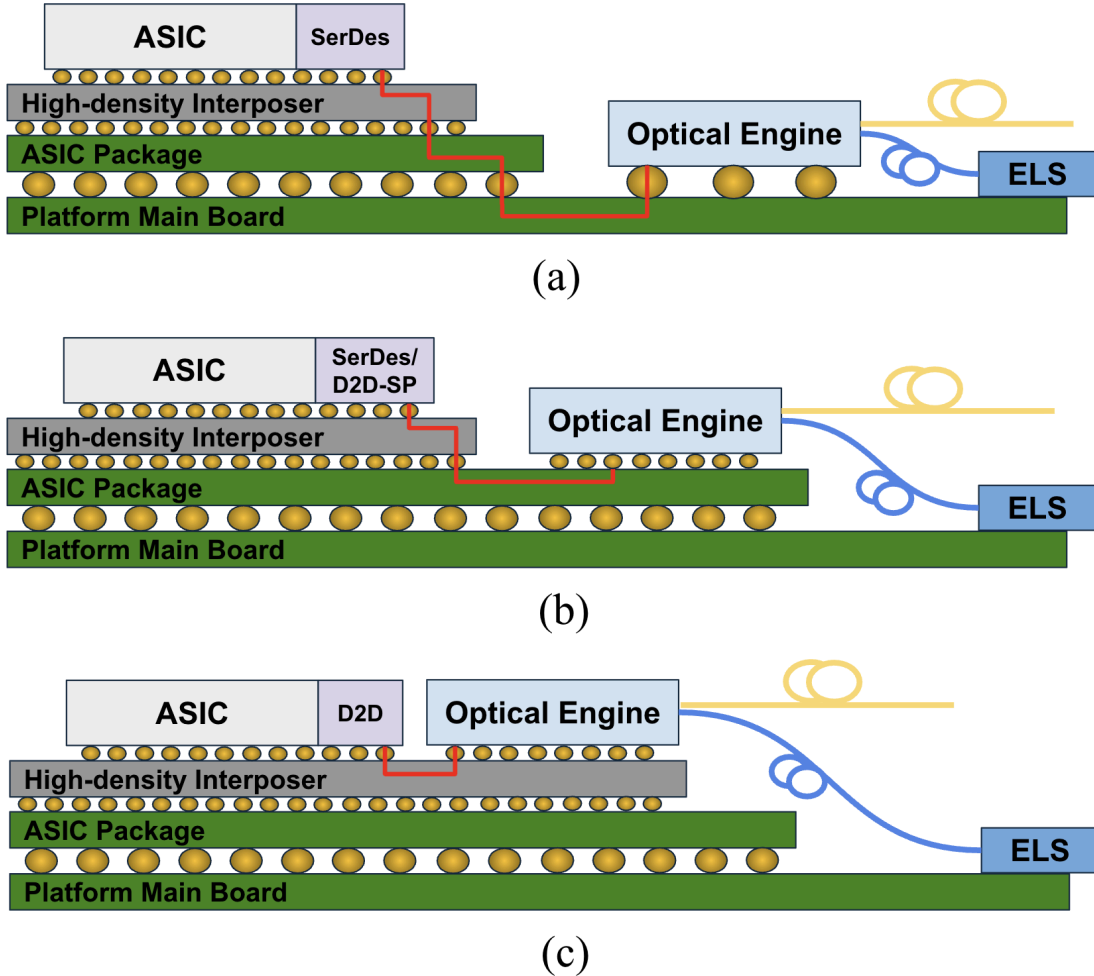


Figure 1. Illustration of three different physical OCI implementations highlighting the different levels of integration with the ASIC; where (a) shows an On-Board Optics (OBO) implementation, (b) shows a package integration implementation and (c) shows an interposer integrated implementation.

The OCI line side interface is based on groups of tightly spaced DWDM channels in the coarse wavelength-division multiplexing (CWDM) grid. To reduce the fiber connectivity and density challenges of the AI scale-up domain, each fiber supports a bidirectional link with both transmit and receive signals counter-propagating on the same fiber. The transmit and receive signals utilize different CWDM wavelength bands. As a result, two wavelength groups of A and B are defined where the group A/B wavelength is transmitted and group B/A wavelength is received for a Type A/B transceiver as depicted in Figure 2. Note that the symmetry in physical

implementation of the OCI transceiver with both Type A and Type B transceivers at both end points is not strictly required as part of this specification. The external laser source (ELS) provides the light to the group A and group B transmitters for modulation.

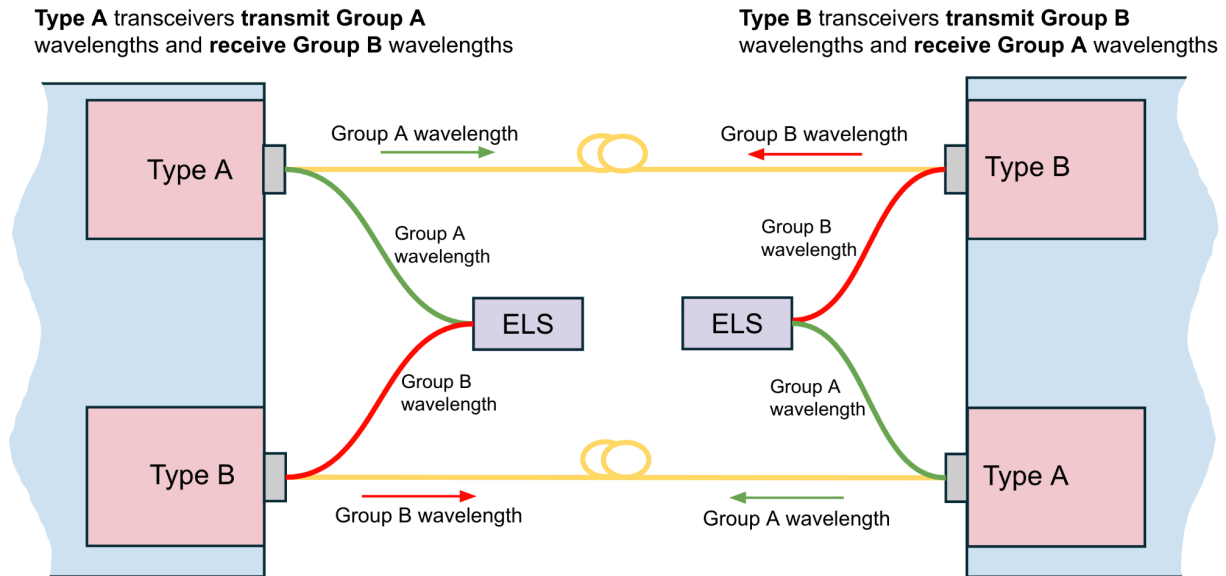


Figure 2. Block diagram of OCI line side connection with group A and group B wavelengths counter-propagating on the same fiber to support a bidirectional link. Type A transceivers transmit Group A wavelengths and receive Group B wavelengths while Type B transceivers transmit Group B wavelengths and receive Group A wavelengths.

Common Terms

The following terms are used in this document.

- OCI – Optical Compute Interconnect
- AI – Artificial Intelligence
- 200GE – 200 Gigabit Ethernet
- 400GE – 400 Gigabit Ethernet
- 800GE – 800 Gigabit Ethernet
- 1.6TE – 1.6 Terabit Ethernet
- CWDM – Coarse Wavelength-Division Multiplexing
- DWDM – Dense Wavelength-Division Multiplexing
- PCS – Physical Coding Sublayer
- PMA – Physical Medium Attachment
- PMD – Physical Medium Dependent
- FEC – Forward Error Correction
- BER – Bit Error Rate
- CDR – Clock Data Recovery
- PRBS – Pseudo Random Binary Sequence
- SSPR – Short Stressed Pattern Random
- ER – Extinction Ratio
- SMSR – Side Mode Suppression Ratio
- OMA – Optical Modulation Amplitude
- AOP – Average Optical Power
- TDEC – Transmitter and Dispersion Eye Closure
- SEC – Stressed Eye Closure
- MUX – Multiplexer
- DEMUX – De-multiplexer
- TX – Transmitter
- RX – Receiver
- NRZ – Non-Return-to-Zero
- CMIS – Common Management Interface Specification
- OE – Optical Engine
- ELS – External Laser Source
- MRR – Micro-Ring Resonator
- ASIC – Application-Specific Integrated Circuit
- D2D – Die-to-Die
- OBO – On-Board Optics
- SMF – Single-Mode Fiber
- PMF – Polarization-Maintaining Fiber
- MPI – Multipath Interference
- IL – Insertion Loss
- CD – Chromatic Dispersion
- RIN – Relative Intensity Noise
- LW – Linewidth
- VDM – Versatile Diagnostic Monitoring

References

The following documents are referenced in generating this specification:

- OIF CMIS revision 5.3 – Common Management Interface Specification
- OIF ELSFP implementation (OIF-ELSFP-01.0)
- IEEE Draft P802.3dj, 2p3

1. Physical Coding Sublayer (PCS) Bit Mapping and Physical Medium Attachment (PMA) Sublayer

Figure 1-1 shows the relationship of the OCI PMA and physical medium dependent (PMD) sublayers to the IEEE 802.3 PCS and PMA sublayers, along with other Ethernet sublayers.

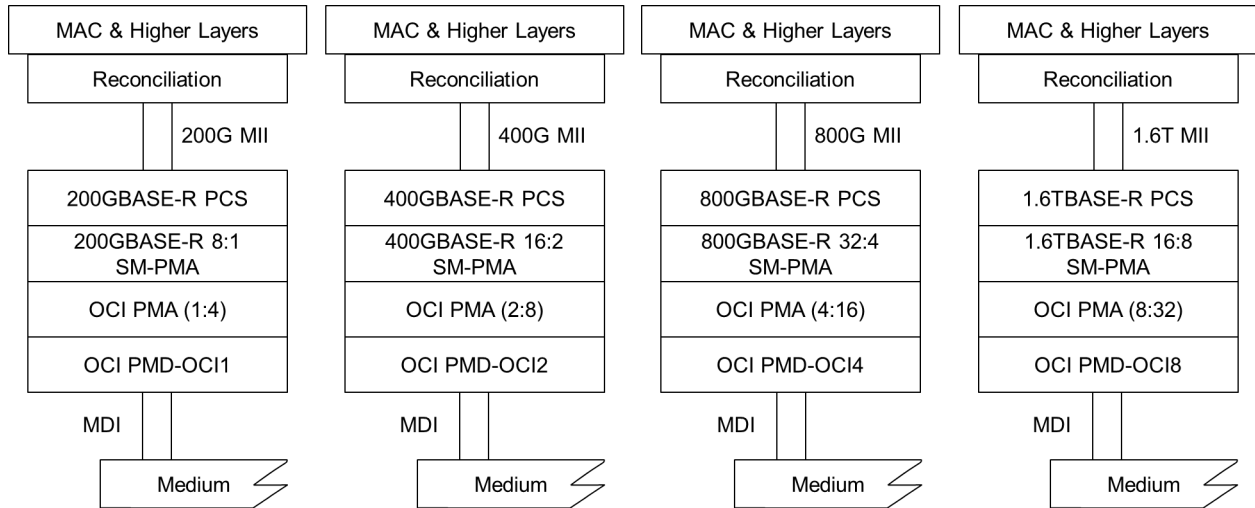


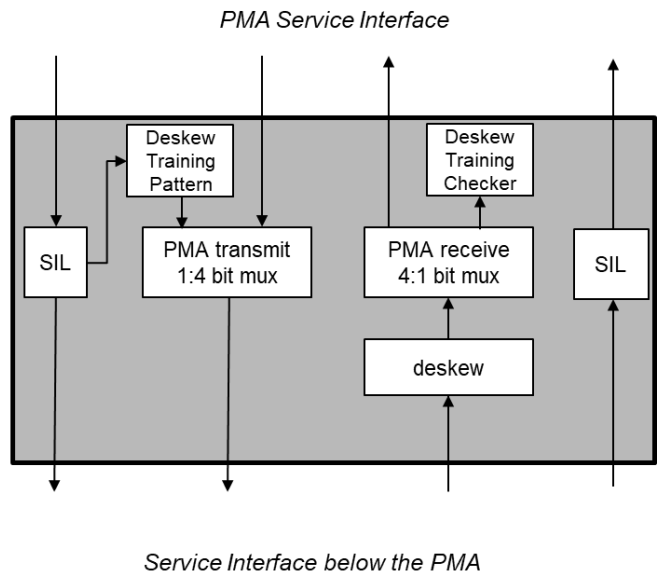
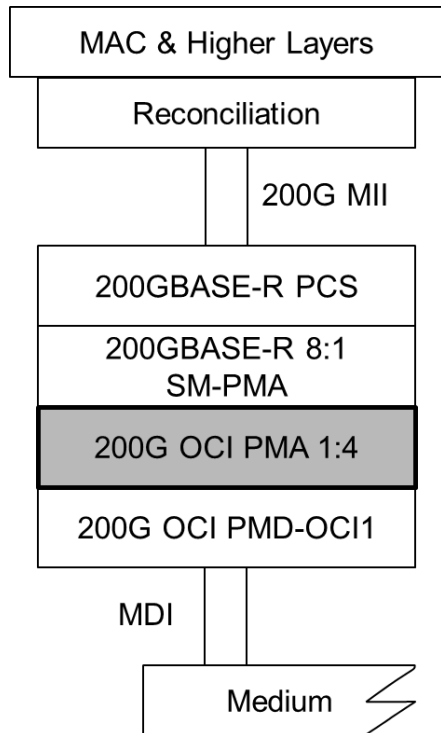
Figure 1-1. Relationship of OCI PMA and PMD to IEEE 802.3 Ethernet model

The OCI PMA interfaces with the 200GBASE-R 8:1 SM-PMA, 400GBASE-R 16:2 SM-PMA, 800GBASE-R 32:4 SM-PMA, and 1.6TBASE-R 16:8 SM-PMA. The OCI PMAs m:n have the variables m and n to denote the number of input and output lanes of each PMA, where:

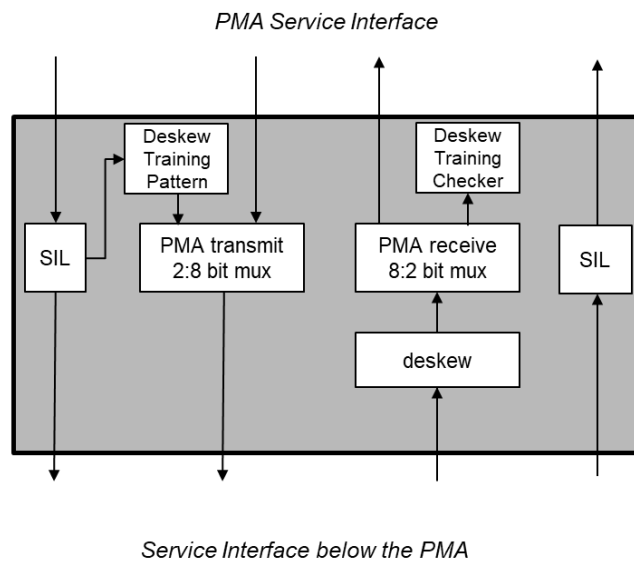
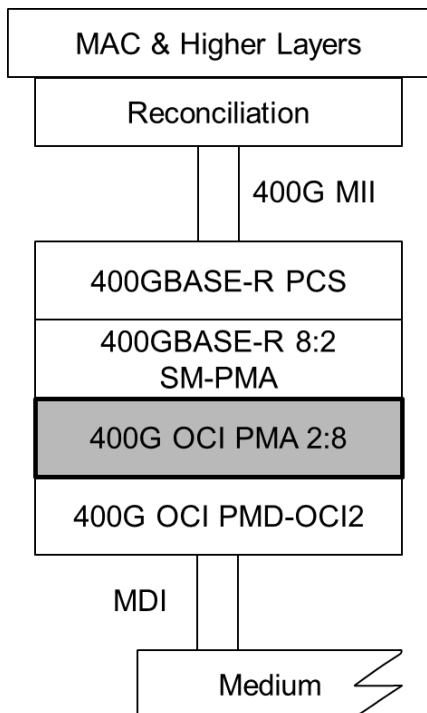
- m is 1 and n is 4 for the 200G OCI 1:4 PMA
- m is 2 and n is 8 for the 400G OCI 2:8 PMA
- m is 4 and n is 16 for the 800G OCI 4:16 PMA
- m is 8 and n is 32 for the 1.6T OCI 8:32 PMA

As illustrated in the shaded box of Figure 1-2 in the transmit (demultiplexing) direction, the m:n PMAs demultiplex the m instances of serial 212.5 Gbps streams to n PMALs of 53.125 Gbps at the service interface below the PMA. In the receive (multiplexing) direction, the m:n PMAs multiplex the n PMALs of 53.125 Gbps at the service interface below the PMA to m PCSs of 212.5 Gbps toward the PMA service interface.

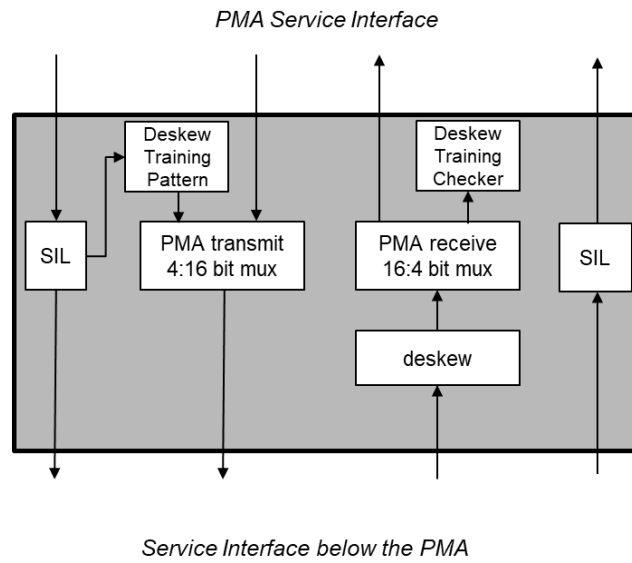
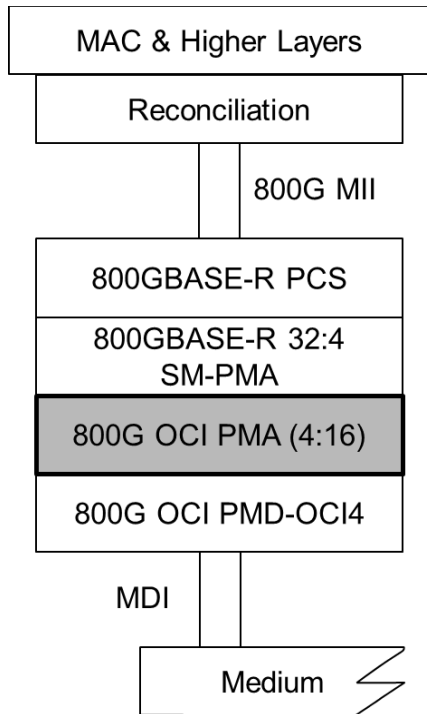
The OCI PMD uses 53.125 Gbaud non-return-to-zero (NRZ) modulation on 4 wavelengths to transport a 212.5 Gbps stream of serial data. The 4 NRZ patterns are deskewed within the OCI-PMD each time a link is established.



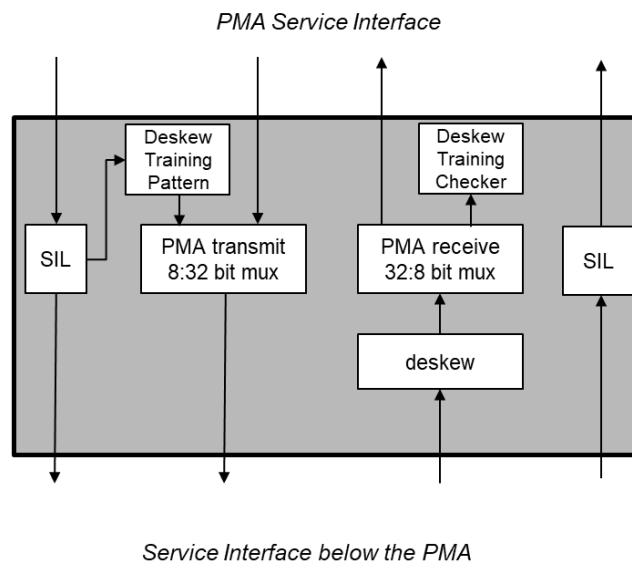
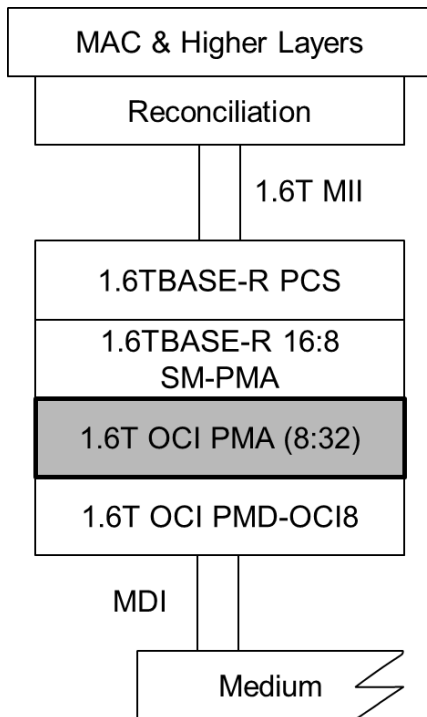
(a)



(b)



(c)



(d)

Figure 1-2. Functional block diagram, (a) 200G OCI 1:4 PMA, (b) 400G OCI 2:8 PMA, (c) 800G OCI 4:16 PMA, and (d) 1.6T OCI 8:32 PMA

1.1. Deskew State Diagram Description

Figure 1-3 and Tables 1-1, 1-2 and 1-3 show the requirements for a state machine which realigns the RX data bits from 4 optical channels within each 212.5 Gbps lane to remove any skew between channels which may be introduced by optical fiber dispersion or other effects. This deskew process occurs in hardware before passing RX data up the protocol stack. Since bidirectional links are used, both ends of a link begin this process after hardware initialization. The deskew state machines remain active indefinitely.

The **Deskew_Data_Relink** state squelches modulation (but not average power) on the four NRZ transmitter channels within each 212.5 Gbps lane in order to signal to the other end of the bidirectional link that it should also begin the deskew process. After TX squelching is applied for `relink_squelch_tx_duration` (see Table 1-3), all 4 transmitters within a 212.5 Gbps lane begin sending the 160 bit deskew training patterns from Table 1-1. Note that all patterns in Tables 1-1 and 1-2 are identical except in bits 23:16. The Data Channel columns of Tables 1-1 and 1-2 refer to the relative wavelength of each channel, with 0 representing the shortest wavelength. Depending on the implementation, physical channel numbers might not correspond to these wavelength indices.

The **Deskew_Data_Detect** state waits until the received data matches the training patterns before proceeding to the **Deskew_Data_Sync** state. If more time than `timeout_data_detect` (Table 1-3) expires without identifying the training patterns, return to the **Deskew_Data_Relink** state. Note that the pattern identification mechanism should be functional at bit error rates (BERs) of up to $1E-4$ and should be able to identify the presence of the pattern successfully.

The **Deskew_Data_Sync** state computes the integer unit interval bit timing skew between the 4 received channels and applies hardware delay to realign the 160 bit patterns. As an example, this can be accomplished by inserting digital delays after phase first in, first outs (FIFOs) for each channel. The earliest arriving channel data can thus be delayed to match the latest arriving data. The range of adjustment must compensate for 0 to 7 unit intervals of relative delay between the earliest and latest arriving 53.125 Gbps NRZ channel data. This provides sufficient range to compensate for chromatic dispersion in the fiber as well as electrical delay mismatch between channels. Note that the pattern skew identification mechanism should be functional at BERs of up to $1E-4$ and should be able to identify the presence of the pattern and the skew between channels successfully. The deskew pattern detection must be robust in the presence of link impairments such as multipath interference (MPI) and back-reflection. If more time than `timeout_data_sync` (Table 1-3) expires without computing the required delay and applying it, return to the **Deskew_Data_Relink** state.

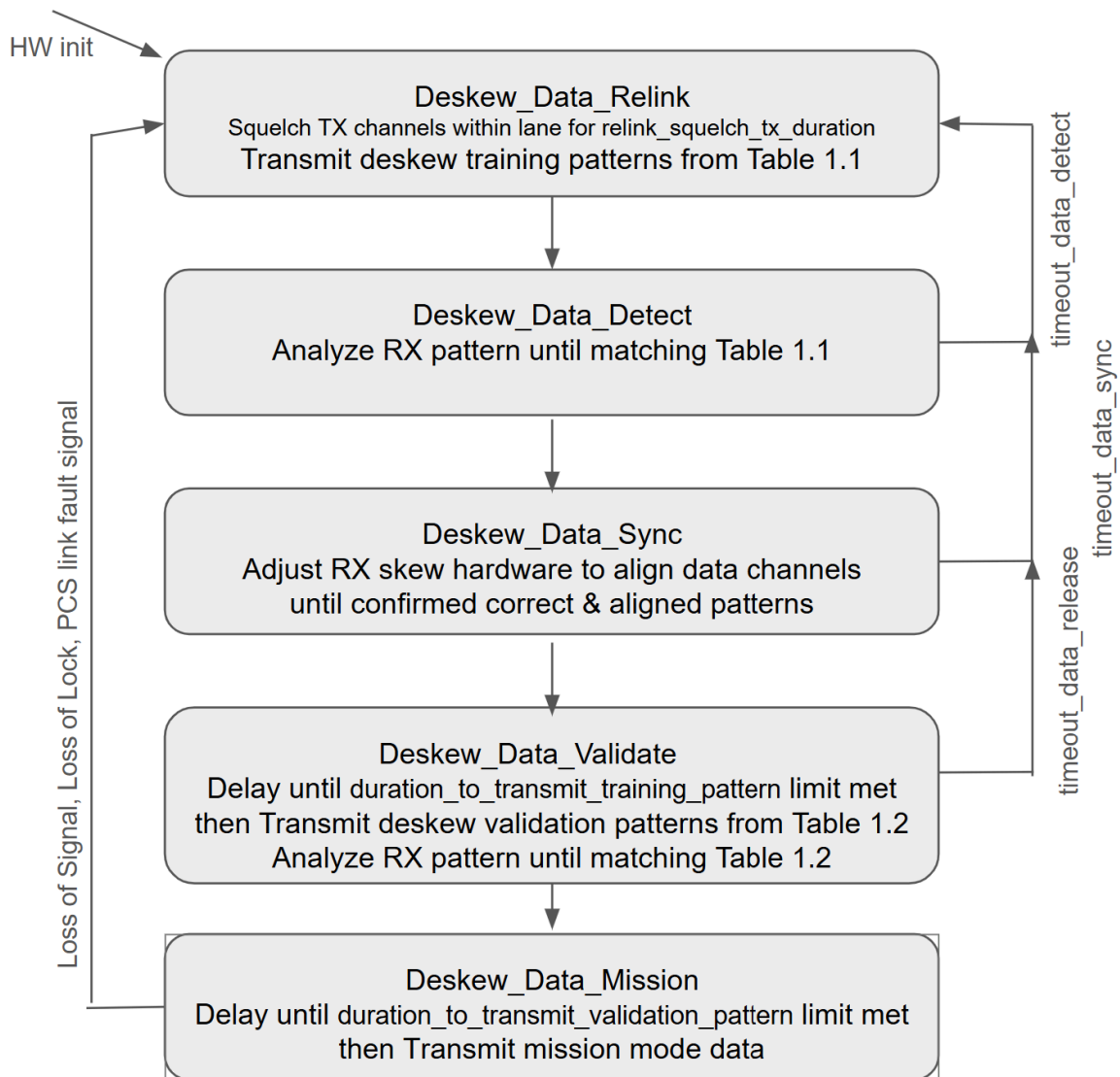


Figure 1-3. Deskew state diagram.

The **Deskew_Data_Validate** state must ensure that the deskew training patterns from Table 1-1 have been applied for at least the time duration_to_transmit_training_pattern limit from Table 1-3. After that time elapses, begin sending the 160 bit deskew release patterns from Table 1-2 on all 4 channel transmitters within a 212.5-Gbps lane. The pattern change must be phase-continuous and glitch-free so that the linked receiver's CDR does not lose lock (which could invalidate the skew removal). After the transmitter pattern changes, begin waiting for the received data to match the release patterns from Table 1-2. Note that the pattern identification mechanism should be functional at BERs of up to 1E-4 and should be able to identify the presence of the pattern successfully.

If more time sending the release pattern than timeout_data_validate (Table 1-3) expires without identifying correctly aligned release patterns, return to the Deskew_Data_Relink state.

The **Deskew_Data_Mission** state must ensure that the deskew release patterns from Table 1-2 have been applied for at least the time duration_to_transmit_release_pattern limit from Table 1-3. After that time elapses, begin sending mission mode data. The pattern change must be phase-continuous and glitch-free so that the linked receiver's CDR does not lose lock (which could invalidate the skew removal). If the RX data becomes invalid, return to the Deskew_Data_Relink state. RX data can be determined invalid based on RX loss of signal, RX loss of clock data recovery (CDR) lock or PCS repeated uncorrectable error detection.

Table 1-1. Deskew training pattern

[illegible]

Table 1-2. Deskew release pattern

[illegible]

Table 1-3. Timing specifications

Timer Name	Min	Target	Max	Unit
relink_squelch_tx_duration	60		75	ms
timeout_data_detect	200		250	ms
timeout_data_sync	100		150	ms
timeout_data_validate	200		450	ms
duration_to_transmit_training_pattern	285			ms
duration_to_transmit_release_pattern	200			ms

Note:

1. It takes no more than 50 ms to detect lack of modulation (t_{loselock}).
2. It takes no more than 50 ms to detect restoration of modulation (t_{lock}).
3. It takes no more than 100 ms to determine whether the incoming patterns match the target (t_{detect}).
4. It takes no more than 100 ms to determine the skew between channels and correct it. (t_{skew}).
5. $\text{duration_to_transmit_training_pattern}$ should be at least $\min(\text{relink_squelch_tx_duration}) - \max(t_{\text{loselock}}) + \max(\text{relink_squelch_tx_duration}) + t_{\text{detect}} + t_{\text{skew}}$ which is equal to 285 ms.
6. Skew between channels at TX optical output < 2 UI from electrical and optical routing.
7. Skew introduced by fiber material and modal chromatic dispersion < 3 UI.
8. Skew between channels introduced after RX fiber coupling due to optical and electrical routing < 2 UI.
9. Subject to change.

1.2. Physical Data Transmission Order and Wavelength Assignment

Every 4 bits provided to the OCI PMA are transmitted with the shortest wavelength TX using the bit from the least significant bit position on the OCI PMA input interface, progressively increasing toward longer wavelength channels for each increase in bit position. Likewise, data bits recovered from the shortest received wavelength channel are delivered to the least significant bit position on the OCI PMA output interface.

2. Optical Specifications

Each optical fiber port shall support 212.5 Gbps data rate bidirectionally with the transmit and receive signals on the same fiber at separate wavelength groups. The 212.5 Gbps electrical signals are formed by aggregating four DWDM 53.125 Gbps NRZ optical signals as described in the previous section. Figure 2-1 shows the DWDM wavelength channels and CWDM wavelength groups for transmitter and receiver supported by the same fiber. The Type A/B transceivers always transmit group A/B wavelengths which are specified in this section. A bidirectional link connects an A type transceiver on one side to a B type transceiver on the other end. While Figure 2-1 illustrates an example symmetric implementation with both Type A and Type B transceivers at both end points, this symmetry is not strictly required as part of the specification and an asymmetric physical implementation can also be compliant.

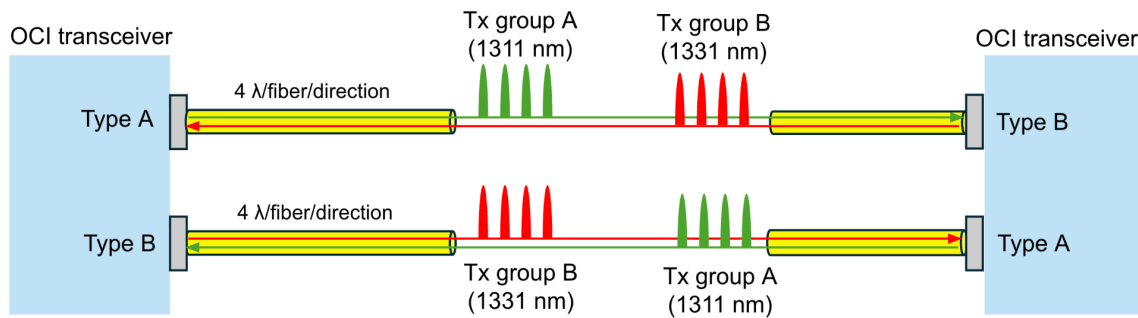


Figure 2-1. Illustration of DWDM wavelength channels and CWDM wavelength groups for a bidirectional link.

2.1. General Characteristics

The general optical specifications are defined in Table 2-1.

Table 2-1. General optical specification

Parameter	Symbol	Min	Typical	Max	Unit
Optical signaling rate, each wavelength			53.125		Gbaud
Modulation format			NRZ		
Number of CWDM wavelength groups per fiber ¹			2		
Number of DWDM wavelength channels per CWDM group ¹			4		
Note:					
1. See Figure 2-1 for illustration of wavelength channels.					

2.2. Optical Transmitter Characteristics

The optical transmitter characteristics are defined in Table 2-2.

Table 2-2. Optical transmitter specification

Parameter	Symbol	Min	Typical	Max	Unit	Test condition
Group A wavelength 0	WL0_TXA	1307.8	1308	1308.2	nm	PRBS13
Group A wavelength 1	WL1_TXA	1310.08	1310.28	1310.48	nm	PRBS13
Group A wavelength 2	WL2_TXA	1312.38	1312.58	1312.78	nm	PRBS13
Group A wavelength 3	WL3_TXA	1314.68	1314.88	1315.08	nm	PRBS13
Group B wavelength 0	WL0_TXB	1327.49	1327.69	1327.89	nm	PRBS13
Group B wavelength 1	WL1_TXB	1329.85	1330.05	1330.25	nm	PRBS13
Group B wavelength 2	WL2_TXB	1332.21	1332.41	1332.61	nm	PRBS13
Group B wavelength 3	WL3_TXB	1334.58	1334.78	1334.98	nm	PRBS13
Side-mode suppression ratio	SMSR	30			dB	
Total average launch power per group	Pavg_total			6	dBm	PRBS13
Average launch power, each channel ¹	Pavg	-8.5		0	dBm	PRBS13
Optical modulation amplitude (OMA), each channel	OMA	max(-5.5, -6.9+TDEC)		-1	dBm	PRBS13
Difference in launch power (OMA) between any two DWDM channels	dOMA			3	dB	PRBS13

Transmitter and dispersion eye closure (TDEC) ²	TDEC			3.4	dB	SSPR
Extinction ratio, each channel	ER	3.5	4.5		dB	PRBS13
$ \text{TDEC}_{\text{SSPR}} - \text{TDEC}_{\text{PRBS13}} ^3$	dTDEC			0.4	dB	SSPR & PRBS13
Launched OMA of squelched transmitter, each channel ⁴	Tsq_channel			-15	dBm	
Transmitter transition time ⁵				17	ps	SSPR
Transmitter over/undershoot				22	%	SSPR
$\text{RIN}_{21.4\text{OMA}}^6$	RIN_OMA			-138	dB/Hz	PRBS13
Optical return loss tolerance	ORL			21.4	dB	
Transmitter data path reflectance ⁷	Tx_data_Ref			-19	dB	
OE laser input reflectance ⁸	OE_Lin_Ref			-26	dB	
Note: 1. Average launch power, each channel (min) is informative and not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance. 2. Measured assuming a reference receiver with 26.5625 GHz bandwidth (BT-4th order), no equalizer, and vertical histograms centered at 0.4 UI and 0.6 UI. Refer to IEEE 802.3 Standard for more details on the measurement method. A pre-FEC BER threshold equal to 2.4E-4 is assumed in calculation of TDEC in all cases. 3. Difference in TDEC measured with SSPR and PRBS13 patterns. 4. When squelching the transmitter modulation, average power should remain constant so that heater lock can be maintained in the micro-rings at the TX and RX. 5. Transition time is measured with the assumption of 20%-80% transition, for the rising and falling edges of the eye diagram respectively. 6. RIN on OMA measured with 21.4 dB return loss. 7. Transmitter reflectance is defined looking into the transmitter from the data path egress and refers to the TX wavelength band. 8. For implementations that have an ELS. Optical engine laser input reflectance is defined looking into the transmitter on the ELS connections.						

2.3. Optical Receiver Characteristics

The optical receiver characteristics are defined in Table 2-3.

Table 2-3. Optical receiver specification

Parameter	Symbol	Min	Typical	Max	Unit	Test condition
Group A wavelength 0	WL0_RXA	1307.8	1308	1308.2	nm	PRBS13
Group A wavelength 1	WL1_RXA	1310.08	1310.28	1310.48	nm	PRBS13
Group A wavelength 2	WL2_RXA	1312.38	1312.58	1312.78	nm	PRBS13

Group A wavelength 3	WL3_RXA	1314.68	1314.88	1315.08	nm	PRBS13
Group B wavelength 0	WL0_RXB	1327.49	1327.69	1327.89	nm	PRBS13
Group B wavelength 1	WL1_RXB	1329.85	1330.05	1330.25	nm	PRBS13
Group B wavelength 2	WL2_RXB	1332.21	1332.41	1332.61	nm	PRBS13
Group B wavelength 3	WL3_RXB	1334.58	1334.78	1334.98	nm	PRBS13
Damage threshold, each channel		4.5			dBm	
Average receive power, each channel ¹	Pavg	-11		0	dBm	
Difference in receive power (Rx OMA) between any two channels	dOMA			3	dB	
Receiver reflectance	Rx_Ref			-19	dB	
Optical modulation amplitude, each channel	OMA			-1	dBm	
Receiver sensitivity (OMA), each channel	RxSens			max(-8.2, -9.6 +TDEC)	dBm	PRBS31
Stressed receiver sensitivity (OMA), each channel ²	SRS			-6.2	dBm	PRBS31
BER floor ³	BER_FL			1E-6		
Loss of signal assert, AOP on each channel	LOS_A	-19	-16.5	-14	dBm	
Loss of signal hysteresis	LOS_H	1	2	3	dB	
Loss of signal de-assert, AOP on each channel ⁴	LOS_D	-18	-14.5	-11	dBm	
Loss of lock detection delay ⁵	t_LOL			50	ms	

Note:

1. Average receive power, each channel (min) is informative and not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.
2. Measured with conformance test signal at TP3 for the BER = 2.4E-4. The stressed conditions are shown in Table 2-4.
3. Measured with a reference transmitter to produce TDEC greater than or equal to 2dB. The BER at receiver must stay within the specified limit over an OMA range of (-8.2 + TDEC) dBm to -1 dBm.
4. Loss of signal de-assert is informative and can be derived from LOS_A and LOS_H.
5. Measured from modulation on/off change to change in loss of lock flag.

Table 2-4. Stressed receiver sensitivity test conditions

Parameter	Value	Unit
Stressed eye closure (SEC), channel under test	3.4	dB
OMA of each aggressor channel	-3.2	dBm

2.4. Optical Fiber Link Model

The reference optical fiber link model is based on 500m SMF-28 fiber with 2.5 dB total insertion loss mainly driven by the connector losses in the link. It represents the worst link inside the data center backend network that OCI transceivers need to close. The total link loss and dispersion budget are shown in Table 2-5.

Table 2-5. Fiber link parameters

Parameter	Symbol	Min	Typical	Max	Unit	Note
Total fiber link loss	IL			2.5	dB	Dependent on the number of connectors and the connector type
Chromatic dispersion	CD	-0.9		1.7	ps/nm	
MPI penalty tolerance				0.2	dB	

2.5. External Laser Source

The OCI implementation shall use an ELS in compliance with the OIF ELSFP implementation and with the specifications provided in Table 2-6. Light must be coupled into a polarization-maintaining fiber (PMF), which is subsequently connected to the OCI chiplet. Careful consideration of the loss budget and tolerance is essential to ensure ELS can provide adequate output power. The laser source will provide a small control range as a means to tune the optical power on the transmit side, allowing for attenuation from the maximum power level. The optical power levels and control range are Implementation-Specific (IS) and are not specified in Table 2-6.

Table 2-6. ELS optical specifications

Parameter	Symbol	Min	Typical	Max	Unit
Group A wavelength 0	WL0_A	1307.8	1308	1308.2	nm
Group A wavelength 1	WL1_A	1310.08	1310.28	1310.48	nm
Group A wavelength 2	WL2_A	1312.38	1312.58	1312.78	nm
Group A wavelength 3	WL3_A	1314.68	1314.88	1315.08	nm
Group B wavelength 0	WL0_B	1327.49	1327.69	1327.89	nm
Group B wavelength 1	WL1_B	1329.85	1330.05	1330.25	nm
Group B wavelength 2	WL2_B	1332.21	1332.41	1332.61	nm
Group B wavelength 3	WL3_B	1334.58	1334.78	1334.98	nm
Maximum optical power per wavelength ¹	P_ELS_WL	IS		IS	dBm
Maximum optical power per fiber ¹	P_ELS_Fiber	IS		IS	dBm
Optical power control range ¹	ΔP_{ELS}	IS		IS	dB
Optical power in start-up mode per wavelength ²	P_ELS_Start	IS		IS	dBm
Laser RIN ³	RINc	-144			dB/Hz

Laser linewidth	LW			1	MHz
Laser SMSR	SMSR	30			dB
Polarization Extinction Ratio	r_{ex}	16			dB
Output reflectance	ELS_Ref			-26	dB
Optical return loss tolerance				-26	dB
Notes: 1. The ELS features the ability to control the optical power delivered to the OCI module. Two specs govern this power. First, if the optical module requests the maximum possible power from the ELS for a given wavelength, the delivered power is defined by P_ELS_WL. Second, the optical module can request that the ELS power be attenuated at the source. The maximum attenuation which can be requested is defined by the specification for ΔP_{ELS}. This attenuation can be used to manage loss variation within the optical module, along with the variation in the maximum power received from the ELS. 2. “Start-up” mode is primarily for verification of optical continuity. As such, other ELS operating specs, including wavelength, may not be met. 3. Measured at maximum optical return loss.					

3. Optical Engine Management Interface

The transceiver should present a CMIS 5.3-based management interface to the host. This shall be accomplished with a processing core. More details on the hardware and software implementation may be added in the subsequent versions of this document.

Diagnostic features such as electrical and optical loopback modes as well as MPI detection are strongly desired to aid in debugging and self-test.

The optical engine relies on a host layer software implementation which follows CMIS specifications. This chapter introduces the OCI-specific portions within CMIS specifications.

Table 3-1. Specific customizations (for interposer OCI, host layer management might need to support additional banks)

Page	Byte	Bit	Field Name	Field Description	Type
Lower	4	7	FlagsSummaryBank4Page2Ch	1b: at least one Flag is set on Bank 4, Page 2Ch	RO Adv.
		6	FlagsSummaryBank4Page14h	1b: at least one Flag is set on Bank 4, Page 14h	RO Adv.
		5	FlagsSummaryBank4Page12h	1b: at least one Flag is set on Bank 4, Page 12h	RO Adv.
		4	FlagsSummaryBank4Page11h	1b: at least one Flag is set on Bank 4, Page 11h	RO Rqd.
		3	FlagsSummaryBank0Page2Ch	1b: at least one Flag is set on Bank 0, Page 2Ch	RO Adv.
		2	FlagsSummaryBank0Page14h	1b: at least one Flag is set on Bank 0, Page 14h	RO Adv.
		1	FlagsSummaryBank0Page12h	1b: at least one Flag is set on Bank 0, Page 12h	RO Adv.
		0	FlagsSummaryBank0Page11h	1b: at least one Flag is set on Bank 0, Page 11h	RO Rqd.
Lower	5	7	FlagsSummaryBank5Page2Ch	1b: at least one Flag is set on Bank 5, Page 2Ch	RO Adv.
		6	FlagsSummaryBank5Page14h	1b: at least one Flag is set on Bank 5, Page 14h	RO Adv.
		5	FlagsSummaryBank5Page12h	1b: at least one Flag is set on Bank 5, Page 12h	RO Adv.
		4	FlagsSummaryBank5Page11h	1b: at least one Flag is set on Bank 5, Page 11h	RO Rqd.

		3	FlagsSummaryBank1Page2Ch	1b: at least one Flag is set on Bank 1, Page 2Ch	RO Adv.
		2	FlagsSummaryBank1Page14h	1b: at least one Flag is set on Bank 1, Page 14h	RO Adv.
		1	FlagsSummaryBank1Page12h	1b: at least one Flag is set on Bank 1, Page 12h	RO Adv.
		0	FlagsSummaryBank1Page11h	1b: at least one Flag is set on Bank 1, Page 11h	RO Rqd.
Lower	6	7	FlagsSummaryBank6Page2Ch	1b: at least one Flag is set on Bank 6, Page 2Ch	RO Adv.
		6	FlagsSummaryBank6Page14h	1b: at least one Flag is set on Bank 6, Page 14h	RO Adv.
		5	FlagsSummaryBank6Page12h	1b: at least one Flag is set on Bank 6, Page 12h	RO Adv.
		4	FlagsSummaryBank6Page11h	1b: at least one Flag is set on Bank 6, Page 11h	RO Rqd.
		3	FlagsSummaryBank2Page2Ch	1b: at least one Flag is set on Bank 2, Page 2Ch	RO Adv.
		2	FlagsSummaryBank2Page14h	1b: at least one Flag is set on Bank 2, Page 14h	RO Adv.
		1	FlagsSummaryBank2Page12h	1b: at least one Flag is set on Bank 2, Page 12h	RO Adv.
		0	FlagsSummaryBank2Page11h	1b: at least one Flag is set on Bank 2, Page 11h	RO Rqd.
Lower	7	7	FlagsSummaryBank7Page2Ch	1b: at least one Flag is set on Bank 7, Page 2Ch	RO Adv.
		6	FlagsSummaryBank7Page14h	1b: at least one Flag is set on Bank 7, Page 14h	RO Adv.
		5	FlagsSummaryBank7Page12h	1b: at least one Flag is set on Bank 7, Page 12h	RO Adv.
		4	FlagsSummaryBank7Page11h	1b: at least one Flag is set on Bank 7, Page 11h	RO Rqd.
		3	FlagsSummaryBank3Page2Ch	1b: at least one Flag is set on Bank 3, Page 2Ch	RO Adv.
		2	FlagsSummaryBank3Page14h	1b: at least one Flag is set on Bank 3, Page 14h	RO Adv.
		1	FlagsSummaryBank3Page12h	1b: at least one Flag is set on Bank 3, Page 12h	RO Adv.
		0	FlagsSummaryBank3Page11h	1b: at least one Flag is set on Bank 3, Page 11h	RO Rqd.
01h	142	7	NetworkPathPagesSupported	Page 16h and NP-related parts of Page 17h supported	RO Rqd.
		6	VDMPagesSupported	VDM Pages 20h-2Fh (partially) supported	RO Rqd.
		5	DiagnosticPagesSupported	Banked Page 13h-14h supported	RO Rqd.
		4	-	Reserved	RO Rqd.
		3	-	Reserved	RO Rqd.
		2	Page03hSupported	User Page 03h supported	RO Rqd.
		1-0	BanksSupported	Banks supported for Pages 10h-1Fh 00b: Bank 0 supported (8 lanes) 01b: Banks 0 and 1 supported (16 lanes) 10b: Banks 0-3 supported (32 lanes) 11b: Banks 0-7 supported (64 lanes)	RO Rqd.

3.1. Flag Summary

CMIS 5.3 only defines flag summaries in the Lower Page Register for banks 0-3. The upper nibble of these registers should be extended to support flag summaries for banks 4-7 if the host implementation uses one host layer to manage 8 optical engines.

3.2. Flight Data Recorder

To facilitate retrieval of debugging information beyond that which is available by the standard CMIS interface, there should be a mechanism for a flight data recorder. CDB protocol definitions could be utilized for this feature. Implementation is proprietary and vendor-specific.

3.3. Versatile Diagnostic Monitoring (VDM) Definitions

VDM provides more monitoring resources outside CMIS page 11h and 14h. The detailed information could be retrieved from VDM specifications (see Table 3-2 for custom observables).

VDM alarms and warnings are specified in section 5 and the thresholds are TBD. Note that the validity of certain VDM observables listed in Table 3-2 depends on the specific physical implementation of the OCI transceiver and should be agreed upon between suppliers and users.

Table 3-2. Custom VDM observables.

Type ID	Monitor resource	VDM					
		Groups	Descriptor	Sample	A/W Thres	A/W Flags	A/W Masks
8	LTP Host	Group 1	20h	24h	28h	2Ch	2Dh
15	PreFEC BER Host						
16	PreFEC BER Line						
100	Vcc Aux						
101	Vcc Aux						
102	Vcc Aux						
103	Vcc Aux						
104	Vcc Aux						
105	Vcc Aux						
106	TX Temperature						
107	RX Temperature						
108	Chiplet Temperature						
109	Channel 0 Tx DMI bias	Group 2	21h	25h	29h		
110	Channel 1 Tx DMI bias						
111	Channel 2 Tx DMI bias						
112	Channel 3 Tx DMI bias						
113	Channel 0 Tx DMI power						
114	Channel 1 Tx DMI power						
115	Channel 2 Tx DMI power						
116	Channel 3 Tx DMI power						
117	Channel 0 Rx DMI power	Group 3	22h	26h	2Ah		
118	Channel 1 Rx DMI power						
119	Channel 2 Rx DMI power						
120	Channel 3 Rx DMI power						

121	Channel 0 PRBS checker BER						
122	Channel 1 PRBS checker BER						
123	Channel 2 PRBS checker BER						
124	Channel 3 PRBS checker BER						
125	Channel 0 MPI metrics	Group 4	23h	27h	2Bh		
126	Channel 1 MPI metrics						
127	Channel 2 MPI metrics						
128	Channel 3 MPI metrics						

4. Digital Diagnostics Specifications

Table 4-1. Digital diagnostics specification

Parameter	Symbol	Min	Max	Unit	Notes
Reported OCI chiplet temperature accuracy ¹	Toci_Err	-3	3	°C	
Reported ELS case temperature accuracy	Telscase_Err	-3	3	°C	
Reported voltage accuracy	Vcc_Err	Vtyp-2%	Vtyp+2%	mV	
Reported Tx output power accuracy ²	Pout_Err	-2	2	dB	
Reported Rx input power accuracy ²	Pin_Err	-2	2	dB	
Reported Tx bias current accuracy ³	Ibias_Err	-10	10	%	
Notes: 1. As measured at the thermal interface between the OCI chiplet and the cold plate above the hottest portion of the OCI chiplet. 2. Referenced to the fiber connection point at the blade or chassis interface 3. Relayed from the ELS					

5. Alarms and Warnings

Alarms and warnings should be reported for each VDM instance, per bank and per lane or channel if applicable. Each of the four DWDM optical wavelength channels has its own alarm and warning reports.

The following is a list of alarm and warning monitoring resources down to channel granularity. Each monitoring resource in this group is per DWDM optical wavelength channel.

- Transmitter bias current (coming from ELS bias current reporting)
- Transmitter optical power
- Receiver optical power
- Optical receiver PRBS checker BER
- Optical receiver MPI metrics

The following is a list of alarm and warning monitoring resources down to data path lane granularity. Each monitoring resource in this group is per 212.5 Gbps electrical data path lane.

- PreFEC BER on the host side
- PreFEC BER on the line side
- LTP on the host side

The following is a list of alarm and warning monitoring resources down to single die granularity. Each monitoring resource in this group is per 1.6 Tbps chiplet.

- Vcc Auxiliary
- TX Temperature
- RX Temperature
- Chiplet Temperature